

PROVISIONAL PATENT APPLICATION

Title

Symmetry-Constrained Boolean Computation Using Spatial State Machines

Field of the Invention

The present invention relates to computation, logic systems, information processing, and spatial computing architectures. More specifically, it relates to a computational logic system in which Boolean-complete operations are embedded within and constrained by spatial symmetry, enabling computation through symmetry-governed state transitions rather than unconstrained symbolic logic.

Background of the Invention

Modern digital computers are built on Boolean logic, typically implemented using logic gates such as NAND, NOR, and IF-THEN constructs. These systems treat bits as independent symbolic values and allow arbitrary Boolean functions, constrained only by circuit design.

Such architectures rely implicitly on a Euclidean spatial substrate in which volumetric points are modeled as cubes arranged in fixed adjacency relationships along orthogonal axes. This assumption enforces fixed neighbor counts, fixed orientations, and global coordinate inheritance, thereby embedding a rigid and implicit logic into the structure of space itself.

While Boolean logic is sufficient for many forms of symbolic computation, it is under-constrained for modeling systems that depend on physical stability, spatial coherence, symmetry preservation, and constrained state evolution, such as biological, chemical, and crystallographic systems.

There is therefore a need for a computational system that:

- remains Boolean-complete,
 - enforces symmetry-based constraints on allowable state transitions,
 - encodes bits relationally rather than axially,
 - and enables computation through spatial reconfiguration governed by symmetry.
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Summary of the Invention

The present invention discloses a **symmetry-constrained computational logic system** in which Boolean-complete states are embedded within spatially defined state machines governed by symmetry groups.

In one embodiment, a fundamental computational unit (a “volumetric logic point”) is represented as a polyhedral structure, such as a tetrahedron or dodecahedron, whose internal states correspond to discrete symmetry-allowed configurations.

A tetrahedral embodiment encodes exactly four states, equivalent to two Boolean bits, where the bits represent relational properties such as spin and chirality rather than Cartesian coordinates. State transitions are not arbitrary Boolean functions but are constrained by the rotational symmetry group of the polyhedron.

Inputs to the system are not symbolic truth values but symmetry-defined operators (e.g., rotations or transformations) that deterministically map one allowed state to another. Invalid or unstable transitions are structurally impossible.

In higher-order embodiments, multiple volumetric logic points interact by rearranging their adjacency relationships in response to global symmetry constraints, thereby enabling computation through coordinated spatial reconfiguration.

This system is Boolean-complete yet symmetry-governed, enabling robust computation without reliance on fixed coordinates, global axes, or unconstrained logic gates.

Brief Description of the Drawings (Optional)

(Figures are optional for a provisional application.)

- **Figure 1:** Tetrahedral state space encoding four Boolean-complete states
- **Figure 2:** Mapping of two relational bits (spin, chirality) to tetrahedral vertices
- **Figure 3:** Symmetry-constrained state transitions under rotational operators
- **Figure 4:** Aggregation of volumetric logic points into a reconfigurable spatial computing lattice

Detailed Description of the Invention

1. Volumetric Logic Point

A volumetric logic point is a discrete computational unit represented by a polyhedral structure. Unlike conventional bits, which are abstract symbols, a volumetric logic point has an internal state space defined by the symmetry of the polyhedron.

In a preferred embodiment, the polyhedron is a **tetrahedron**, which possesses four vertices and rotational symmetry. Each vertex represents a distinct allowed state.

2. Boolean Reduction of the Tetrahedral State Space

The four tetrahedral states correspond exactly to two Boolean bits:

$$\text{State} = (b_1, b_2)$$

where:

- **b₁** represents a spin or orientation bit, and
- **b₂** represents a chirality or handedness bit.

These bits are **relational**, not axial. They do not correspond to orthogonal spatial directions but to symmetry-defined properties invariant under allowed transformations.

Thus, the system is Boolean-complete while avoiding the independence assumptions of standard Boolean logic.

3. Symmetry-Constrained State Transitions

State transitions occur through the application of symmetry-defined operators, such as rotations or transformations associated with the symmetry group of the polyhedron.

Given:

- a current state (one of four tetrahedral states), and
- an input operator selected from a finite set of symmetry-allowed operations,

the system deterministically transitions to exactly one other allowed state.

Importantly:

- Not all Boolean functions are permitted.
- Only transitions consistent with the symmetry group exist.
- Invalid transitions are structurally impossible.

This defines a **symmetry gate**, distinct from NAND or IF–THEN gates.

4. Extension to Higher-Order State Spaces

In further embodiments, the tetrahedral logic point is embedded within a higher-order polyhedral structure, such as a dodecahedron, which constrains a larger finite state space (e.g., 240 tetrahedral configurations).

Each volumetric logic point may:

- internally update its state according to local symmetry rules, and
- externally rearrange its neighbor relationships according to global symmetry constraints.

This two-phase process enables computation through spatial reconfiguration rather than fixed wiring.

5. Reconfigurable Spatial Computation

Unlike cubic lattices with fixed adjacency, volumetric logic points in the present invention may dynamically rearrange their neighbors. Computation occurs through:

1. Local symmetry-constrained state transitions.
2. Global rearrangement to restore or maintain symmetry consistency.

This enables computation without wires, clocks, or global coordinate frames.

Advantages of the Invention

- Boolean-complete without unconstrained logic
 - Built-in error prevention via impossible states
 - Naturally preserves chirality and symmetry
 - Scales through geometry rather than wiring
 - Applicable to biological, chemical, and AI systems
 - Compatible with, but not limited to, digital hardware implementations
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Potential Applications

- Novel computing architectures
 - Spatial or geometric processors
 - Biological and molecular modeling
 - AI systems with symmetry-based learning
 - Quantum-inspired or post-Boolean computation
 - Educational and demonstrative computing devices
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Statement of Inventive Concept

The inventive concept lies in recognizing that Boolean logic can be fully embedded within spatially symmetric state machines, and that constraining Boolean transitions by symmetry yields a fundamentally new and more robust form of computation.

Provisional Nature

This application is submitted as a provisional patent application under 35 U.S.C. § 111(b). No claim is made to any particular implementation, material, scale, or hardware embodiment at this time. All variations consistent with the described principles are intended to be covered by subsequent non-provisional filings.